CHAPTER 5

Digital Ins and Outs

So far in our discussion of programming the Arduino microcontroller, we have generally glossed over the physical interactions of turning things on and off and sensing the world around us that make programming microcontrollers so much fun. Now it's time to fix that. Programming a microcontroller can be rather unique because of its ability to easily turn on and off motors, lights, and other actuators, as well as read from switches and sensors. This is possible on the Arduino Uno by using its 20 input and output (I/O) pins, each having different functions that can be configured in software—including input, output, analog to digital conversion (ADC), and analog output, more accurately called pulse width modulation (PWM)). This chapter will take as its focus the various digital input and output functions available and how to interface the needed hardware to make them work.

What unifies the functions and code in this chapter as being digital is that they all have only two different states: on or off. We have actually already used the Arduino's digital I/O from the very beginning of this book and briefly introduced digital input in the last chapter using a very simple form of a digital sensor in the basic rolling ball tilt switch. Other forms of digital input include pushbuttons, mat switches, magnetic or reed switches, thermostats, passive infrared motion detectors, and oh so many more. We can use digital output to turn on not only LEDs but also to make sound, create movement with motors or solenoids, and even control larger devices using transistors and relays.

We will start with an introduction of the electrical properties of the Arduino's I/O pins, followed by a new project called Noisy Cricket, and then we will move into a thorough examination of the digital functions and how we can incorporate them into our code.

What's needed for this chapter:

- Arduino Uno
- Passive infrared motion detector
- Piezoelectric speaker
- Momentary pushbutton or switch
- 10 kilohm ¼ watt resistor or similar
- Hookup wires
- Solderless breadboard